



OS MIDI Controller – Web Interface

Beautiful dark-themed web interface for configuring your OS MIDI Controller via Web Serial API.

Features

✨ Modern Dark Theme

- Gradient backgrounds
- Smooth animations
- Responsive design

🔌 Web Serial Connection

- Direct serial communication from browser
- Real-time console output
- Send commands to device

🕒 Control Mappings

- Configure 7 buttons (MIDI notes 0-127)
- Configure 4 potentiometers (CC or Pitch Bend)
- Instant apply with "Set" buttons
- Visual feedback

💾 Configuration Management

- Save to EEPROM
- Load from EEPROM
- Reset to factory defaults
- Load current mapping view

🖥️ Live Console

- See all device responses
- Color-coded messages
- Timestamps
- Auto-scroll

Requirements

- **Chrome** or **Edge** browser (Web Serial API support)
- **OS MIDI Controller** connected via USB

How to Use

1. Open the Interface

Simply open `controller.html` in Chrome or Edge:

```
# macOS
open controller.html

# Or drag the file into Chrome
```

2. Connect to Device

1. Click "Connect Serial"
2. Select your device from the popup (look for "usbmodem" on macOS)
3. Status will show "Connected" in green

3. Configure Mappings

Buttons

- Enter MIDI note (0-127) for each button
- Click "Set" to apply
- Notes are sent to device but not saved yet

Potentiometers

- Choose "CC" for Control Change or "PB" for Pitch Bend
- If CC: enter CC number (0-127)
- Click "Set" to apply

4. Save Configuration

After making changes:

1. Click " Save to EEPROM" to persist changes
2. Device will write to EEPROM and confirm

5. Load/Reset

-  Load from EEPROM: Reload saved config from device
-  Reset to Defaults: Restore factory settings
- Load from Device (/m): View current mappings in console

Keyboard Shortcuts

None yet, but coming soon!

Console Commands

The interface sends these commands to the device:

Command	Description
/b <1-7> <0-127>	Map button to MIDI note
/p <1-4> cc <0-127>	Map pot to CC
/p <1-4> pb	Map pot to Pitch Bend
/save	Save to EEPROM
/load	Load from EEPROM
/reset	Reset to defaults
/m	Show current mappings

Default Mappings

Buttons (Drum Kit)

Button	Default Note	Instrument
1	36	Kick Drum
2	38	Snare
3	42	Closed Hi-Hat
4	46	Open Hi-Hat
5	37	Side Stick
6	39	Clap
7	49	Crash Cymbal

Potentiometers

Pot	Default	Description
1	CC 1	Modulation Wheel
2	CC 2	Breath Controller
3	CC 74	Cutoff/Brightness

Pot	Default	Description
4	CC 71	Resonance/Timbre

Troubleshooting

"Web Serial API not supported"

- **Solution:** Use Chrome (v89+) or Edge (v89+)
- Firefox and Safari don't support Web Serial yet

Can't see device in port list

- **Solution:**
 - Make sure device is plugged in
 - Check device shows up in system (macOS: `/dev/cu.usbmodem*`)
 - Try unplugging and replugging
 - Close other apps using the serial port

Connected but no response

- **Solution:**
 - Check baud rate is 115200 (automatic in this interface)
 - Try disconnecting and reconnecting
 - Reset the device

Changes not persisting after reboot

- **Solution:** Remember to click " Save to EEPROM" after making changes!

Advanced Usage

Send Custom Commands

Open browser console (F12) and use:

```
await sendCommand('/your_command');
```

Monitor Serial Data

All received data is logged to the console area with timestamps.

Browser Console (Developer Tools)

Press F12 to open developer console for debugging:

- Network errors
- Serial port issues
- JavaScript errors

Security Notes

- **HTTPS Required:** Web Serial API requires HTTPS in production
- **User Permission:** Browser will always ask permission to access serial port
- **Local Files:** Works with `file://` protocol for local development

Future Enhancements

- ☐ Preset management (save/load multiple configs)
- ☐ MIDI monitor (see notes in real-time)
- ☐ Export/import config as JSON
- ☐ Keyboard shortcuts
- ☐ Button test mode (highlight on press)
- ☐ Potentiometer visualization
- ☐ Dark/light theme toggle

Credits

OS MIDI Controller Interface v1.0

Built with vanilla HTML/CSS/JavaScript

Uses Web Serial API

Dark theme with gradient design

Need Help? Check the main project documentation or open an issue.